

Data Scraping, Data Cleaning and Data Insight of Doctor BD Dataset

Lab Report

1. Introduction / Objective

This lab report presents the data cleaning and data insight phase of a project that scraped doctor-directory information from the website doctor.bd. The objective was to clean the raw scraped dataset, verify its structural integrity, and extract meaningful statistical insights regarding the distribution of medical specialties, locations, hospitals, designations, and professional experience among the listed doctors.

2. Dataset Description

The dataset consists of 2,000 doctor records collected from doctor.bd, stored across 8 columns: doctor_name, specialty, qualification, designation, hospital, location, experience, and profile_url.

3. Data Cleaning

The raw dataset was loaded using pandas and inspected for row count, column count, data types, and missing values. Duplicate records were identified and removed based on the unique profile_url field. Extra whitespace was stripped from all text columns, and missing or empty values were standardized as "N/A". The experience column was further cleaned and standardized into a consistent "X+ Years" format wherever a numeric value could be extracted. The original doctors_data.csv was preserved unmodified, and the cleaned dataset was saved separately as cleaned_doctors_data.csv.

Cleaning Metric	Result
Total records	2,000
Total columns	8
Duplicate records removed	0
designation missing	1 (0.1%)
hospital missing	1 (0.1%)
experience missing	1,892 (94.6%)
Other columns	No missing values

Note: the experience column has a very high missing-data rate of 94.6%, since the source website itself does not publish a numeric years-of-experience value for the large majority of listed doctors.

4. Data Insights

Table 1: Top 5 Specialties, Locations, and Hospitals (of the Top 10 analyzed)

Specialty	Count	Location	Count	Hospital	Count
Gynaecology & Obstetrics	273	Dhaka	958	Chattogram MCH	128
Pediatrics	163	Chattogram	250	Dhaka MCH	114
Medicine	159	Sylhet	128	Bangladesh Medical University	109
Cardiology	142	Khulna	113	Sylhet MAG Osmani MCH	64
Orthopedics	116	Cumilla	90	Rajshahi MCH	59

Table 2: Top 5 Designations and Experience Summary

Designation	Count	Experience Metric	Value
Consultant	594	Doctors with experience data	108 / 2,000 (5.4%)
Assistant Professor	476	Minimum experience	2+ Years
Associate Professor	270	Maximum experience	37+ Years
Senior Consultant	101	Average experience	12.8+ Years
Professor & Head	78		

Gynaecology & Obstetrics is the leading specialty, and Dhaka dominates as the primary location, together indicating strong urban concentration of listed practitioners. Consultant is the most common designation, reflecting the hospital-practice-heavy composition of the dataset.

Charts

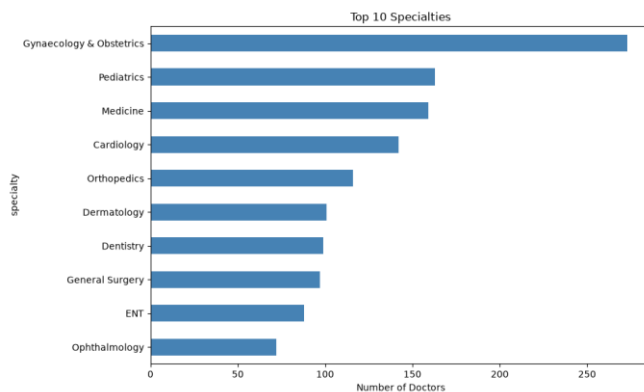


Fig. 1: Top 10 Specialties

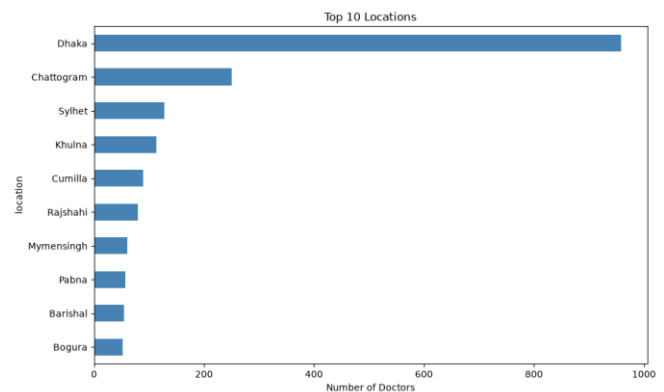


Fig. 2: Top 10 Locations

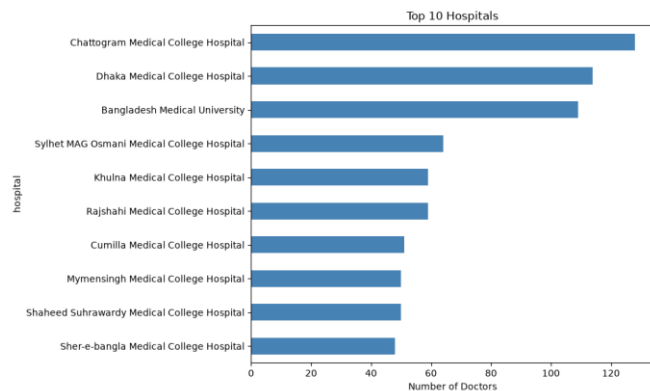


Fig. 3: Top 10 Hospitals

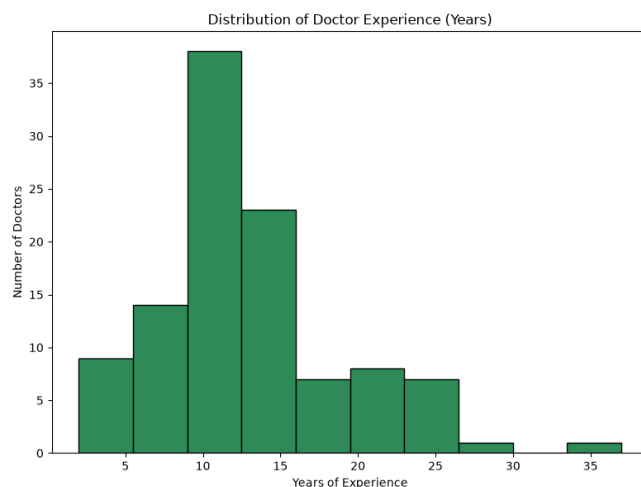


Fig. 4: Experience Distribution

5. Conclusion

The cleaning process produced a fully de-duplicated, standardized dataset of 2,000 doctor records with only negligible missing data in the designation and hospital fields. The main data-quality limitation is the experience column, missing for 94.6% of records due to the source website's own limited reporting. The insights confirm that Gynaecology & Obstetrics, Pediatrics, and Medicine are the leading specialties, that doctor listings are heavily concentrated in Dhaka and Chattogram, and that Consultant is the most common professional designation in the dataset.